Major Milestones and State-of-the-Art Models in AI

1 Computer Vision

1.1 Image Classification

The evolution of image classification began with early convolutional networks like LeNet-5 (1998) and peaked in deep architectures such as AlexNet (2012). Major milestones and the current state-of-the-art models include:

- **1998: LeNet-5**
  - 7 layers: input, 2 convolutional, 2 subsampling (pooling), 2 fully connected, output.
  - Input: 32×32 grayscale image.
  - Conv1: 6 filters of size 5×5, stride 1, output $28 \times 28 \times 6$.
  - Pool1: 2×2 average pooling, output $14 \times 14 \times 6$.
  - Conv2: 16 filters of size 5×5, output $10 \times 10 \times 16$.
  - Pool2: 2×2 average pooling, output $5 \times 5 \times 16$.
  - FC1: 120 units; FC2: 84 units; Output: 10-way softmax.
  - Total parameters: $\sim$60K.
- **2012: AlexNet**
  - 8 layers: 5 convolutional, 3 fully connected.
  - Input: $227 \times 227 \times 3$ RGB image.
  - Conv1: 96 filters of size 11×11, stride 4, output $55 \times 55 \times 96$.
  - Pool1: 3×3 max pooling, output $27 \times 27 \times 96$.
  - Conv2: 256 filters of size 5×5, output $13 \times 13 \times 256$.
  - Pool2: 3×3 max pooling, output $6 \times 6 \times 256$.
  - Conv3-5: deeper layers with ReLU activations and dropout.
  - FC1: 4096 units; FC2: 4096 units; Output: 1000-way softmax.
  - Total parameters: $\sim$60M.
- **2014: VGG-16/19**
  - 16/19 layers: 13/16 convolutional, 3 fully connected.
  - Input: $224 \times 224 \times 3$ RGB image.
  - Conv1-5: 2-3 convolutional layers per block with 3×3 filters, max pooling after each block.
  - FC1: 4096 units; FC2: 4096 units; Output: 1000-way softmax.
  - Total parameters: $\sim$138M (VGG-16).
  - Known for deep architecture and small receptive fields.
- **2014: GoogLeNet (Inception v1)**

  - 22 layers: inception modules with parallel convolutions.
  - Input: $224 \times 224 \times 3$ RGB image.
  - Inception modules with 1×1, 3×3, and 5×5 convolutions.
  - Global average pooling instead of fully connected layers.
  - Total parameters: $\sim$6.8M.
  - Introduced auxiliary classifiers for better gradient flow.
  - Known for inception modules that allow multi-scale feature extraction.

- **2015: ResNet**
  - 152 layers: residual connections to enable very deep networks.
  - Input: $224 \times 224 \times 3$ RGB image.
  - Residual blocks with skip connections to mitigate vanishing gradients.
  - Global average pooling instead of fully connected layers.
  - Total parameters: $\sim$60M (ResNet-152).
  - Known for enabling training of extremely deep networks.

- **2017: DenseNet**
  - 121 layers: dense connections between layers.
  - Input: $224 \times 224 \times 3$ RGB image.
  - Each layer receives feature maps from all previous layers.
  - Global average pooling instead of fully connected layers.
  - Total parameters: $\sim$8M (DenseNet-121).
  - Known for efficient feature reuse and reduced number of parameters.

- **2019: EfficientNet**
  - 8 models (B0-B7): compound scaling of depth, width, and resolution.
  - Input: $224 \times 224 \times 3$ RGB image.
  - Mobile inverted bottleneck convolution (MBConv) blocks.
  - Global average pooling instead of fully connected layers.
  - Total parameters: $\sim$5.3M (EfficientNet-B0) to $\sim$66M (EfficientNet-B7).
  - Known for achieving high accuracy with fewer parameters.

- **2020-2024: ViT, ConvNeXt, Swin Transformer**
  - Vision Transformer (ViT): Transformer architecture applied to image patches.
  - ConvNeXt: Modernized ResNet with better performance on ImageNet.
  - Swin Transformer: Hierarchical Transformer with shifted windows for local and global attention.
  - Known for leveraging self-attention mechanisms for image classification.

The current state-of-the-art (SOTA) in image classification is a combination of powerful convolutional neural networks (CNNs) and transformer models. CNNs like EfficientNet and ResNet continue to be dominant, while Vision Transformers (ViT) are also making significant inroads.

1.2 Image Segmentation

Key milestones in semantic segmentation:

- **2015: FCN** - Fully Convolutional Networks for pixel-wise classification.
  - Replaces fully connected layers with convolutional layers.
  - Uses upsampling to produce dense predictions.
  - Introduced skip connections for better localization.
- **2015: U-Net** - Encoder-decoder architecture for biomedical segmentation.
  - Contracting path (encoder) with downsampling.
  - Expansive path (decoder) with upsampling and skip connections.
  - Known for its effectiveness in small datasets.
- **2016-2017: SegNet, DeepLab v1-v3** - Variants of encoder-decoder architectures.
  - SegNet: Encoder-decoder with max pooling indices for upsampling.
  - DeepLab: Atrous convolution for multi-scale context.
  - DeepLab v3: Atrous Spatial Pyramid Pooling (ASPP) for robust feature extraction.
- **2017: Mask R-CNN** - Extends Faster R-CNN for instance segmentation.
  - Adds a branch for predicting segmentation masks.
  - Uses Region Proposal Networks (RPN) for object detection.
  - Known for its high accuracy in instance segmentation tasks.
- **2018-2024: PSPNet, HRNet, SETR** - Advanced segmentation architectures.
  - PSPNet: Pyramid Scene Parsing Network for global context.
  - HRNet: High-Resolution Network maintaining high-resolution representations.
  - SETR: Transformer-based segmentation with self-attention mechanisms.

The current state-of-the-art in image segmentation includes a mix of CNN-based architectures like U-Net and Mask R-CNN, as well as transformer-based models like SETR and HRNet. These models excel in both semantic and instance segmentation tasks, leveraging advanced techniques such as multi-scale feature extraction and self-attention.

1.3 Object Detection

1.3.1 Two-Stage Detectors:

- **R-CNN (2014):**
  - Uses selective search to generate region proposals.
  - Extracts features from each region using a CNN.
  - Classifies regions with SVMs and refines boxes with regression.
  - Slow due to per-region CNN computation.
- **Fast R-CNN (2015):**
  - Processes the whole image with a CNN once.
  - Uses RoI pooling to extract features for each region proposal.
  - Jointly predicts class and bounding box offsets with a softmax and regression head.
  - Faster and more accurate than R-CNN.

- **Faster R-CNN (2015):**
  - Introduces Region Proposal Network (RPN) to generate proposals directly from feature maps.
  - Shares convolutional features between RPN and detection network.
  - Achieves real-time detection speeds.
- **Mask R-CNN (2017):**
  - Extends Faster R-CNN by adding a parallel branch for predicting segmentation masks.
  - Uses RoIAlign for better mask localization.
  - Supports both object detection and instance segmentation.

1.3.2 One-Stage Detectors:

- **YOLO (v1-v11):**
  - Unified architecture that predicts bounding boxes and class probabilities directly from full images in a single forward pass.
  - Backbone: Darknet (various versions), CSPNet, or Transformer-based backbones in later versions.
  - Known for real-time speed and end-to-end training.
- **SSD (Single Shot MultiBox Detector):**
  - Uses a base network (e.g., VGG-16) with additional convolutional layers for multi-scale feature maps.
  - Predicts category scores and box offsets for a fixed set of default bounding boxes.
  - Efficient and accurate for real-time detection.
- **RetinaNet:**
  - Backbone: ResNet + Feature Pyramid Network (FPN).
  - Introduces focal loss to address class imbalance.
  - Dense detection with high accuracy for small and large objects.
- **DETR (DEtection TRansformer):**
  - Transformer-based end-to-end object detector.
  - Uses a CNN backbone (e.g., ResNet) for feature extraction, followed by a Transformer encoder-decoder.
  - Predicts a fixed set of objects using bipartite matching loss.
  - Simplifies the detection pipeline by removing the need for anchor boxes and NMS.

1.4 Image Generation

From GANs to diffusion models:

- **2014: GAN** - Minimax adversarial loss.
  Architecture: Two networks (generator and discriminator) trained in opposition.
- **2016: DCGAN** - Deep convolutional GAN for stable image generation.
  Architecture: Uses convolutional layers in both generator and discriminator.
- **2017: Pix2Pix, CycleGAN** - Image-to-image translation.
  Architecture: Encoder-decoder (Pix2Pix) and cycle-consistent GANs (CycleGAN) for unpaired translation.
- **2018: BigGAN** - Large-scale GAN for high-fidelity images.
  Architecture: Scaled-up GAN with class conditioning and large batch sizes.

- **2019-2020: StyleGAN1/2** - Style-based generator for controllable synthesis.
  Architecture: Mapping network and adaptive instance normalization for style control.
- **2020-2024: Diffusion Models (DDPM, Stable Diffusion, Imagen)** - Iterative denoising for image synthesis.
  Architecture: Gradually denoise random noise using a U-Net or Transformer backbone.

2 Natural Language Processing

2.1 Sentiment Classification

- **2014: CNN/LSTM-based models (Kim, 2014)** - Early deep learning approaches for text representation and classification.
- **2018: BERT (Google)** - Introduced bidirectional training of Transformers for pre-training and fine-tuning.
- **2019: RoBERTa (Facebook AI)** - Optimized BERT pretraining, improving performance by modifying key hyperparameters.
- **2019: XLNet (Google/CMU)** - Generalized autoregressive pretraining that learns bidirectional contexts.
- **2020-2024: GPT-2/3/4 (OpenAI), T5 (Google), PaLM (Google), Gemini (Google)** - Large-scale autoregressive and encoder-decoder models excelling with fine-tuning and prompting; Gemini offers native multimodality.

2.2 Text Generation

- **2014: Seq2Seq (Google)** - Encoder-decoder architecture for sequence-to-sequence tasks like machine translation.
- **2017: Transformer (Google)** - Introduced self-attention mechanism, becoming the foundation for many subsequent models.
- **2018-2023: GPT-1/2/3/4 (OpenAI)** - Pioneering large-scale autoregressive models for coherent text generation.
- **2019: T5 (Google)** - A unified Transformer model for all NLP tasks, framed as text-to-text problems.
- **2023: LLaMA (Meta)** - A collection of foundational large language models for various research and commercial applications.
- **2022: OPT (Meta)** - Open-source suite of large language models, aiming for broader accessibility in research.
- **2023-2025: Gemini (Google)** - Multimodal models capable of generating text from diverse inputs (text, image, audio, video).
- Autoregressive decoder (e.g., GPT, Gemini) and encoder-decoder (e.g., T5) used for generation.

3 Audio

3.1 Speech Recognition

- **2015: Listen, Attend and Spell (Google)** - An end-to-end neural network for speech transcription, using attention.
- **2015: DeepSpeech (Mozilla/Baidu)** - End-to-end speech recognition using recurrent neural networks and CTC loss.
- **2020: wav2vec 2.0 (Facebook AI)** - Self-supervised learning framework for speech representation learning.
- **2022: Whisper (OpenAI)** - A large Transformer encoder-decoder model for robust speech recognition and transcription.
- **2023-2025: Gemini (Google)** - Multimodal models with strong audio understanding capabilities, including transcription and summarization.

3.2 Text-to-Speech (TTS)

- **2016: WaveNet (DeepMind/Google)** - A groundbreaking generative model for raw audio waveforms, producing highly natural speech.
- **2017: Tacotron (Google)** - An end-to-end speech synthesis model that generates mel spectrograms from text, followed by a vocoder.
- **2017: Tacotron 2 (Google)** - Improved version of Tacotron, combining a sequence-to-sequence model with WaveNet for high-quality speech.
- **2019: FastSpeech (Microsoft)** - A feed-forward network for parallel and faster speech synthesis.
- **2019: MelGAN (NVIDIA)** - A generative adversarial network for high-fidelity and efficient waveform generation from mel spectrograms.
- **2020: HiFi-GAN (Kakao Enterprise)** - A fast and high-fidelity GAN-based vocoder for speech synthesis.
- **2021: VITS (Kakao Enterprise)** - An end-to-end VAE-GAN based TTS model for high-quality and efficient speech synthesis.
- **2024-2025: Gemini (Google)** - Advanced text-to-speech capabilities, including multi-speaker and expressive audio output (preview).

4 Recommender Systems

- **1990s-2000s: Collaborative Filtering** - Recommends items based on user preferences or item similarities.
- **2006: Matrix Factorization** - Decomposes user-item interaction matrices to discover latent features (e.g., Netflix Prize era).
- **2016: Wide & Deep (Google)** - Combines the memorization of wide linear models with the generalization of deep neural networks.
- **2017: NCF (Neural Collaborative Filtering)** - Replaces matrix factorization with a neural network for modeling user-item interactions.
- **2017: DeepFM** - Integrates Factorization Machines with deep neural networks for feature interactions.
- **2017: xDeepFM** - Explicitly models feature interactions at different orders in a compressed interaction network.
- **2018+: SASRec (Self-Attentive Sequential Recommendation)** - Uses Transformer architecture to model sequential user behavior for next-item prediction.
- **2019+: LightGCN (Graph Convolutional Networks)** - Simplifies GCNs for recommendations by only keeping neighborhood aggregation in graph convolutions.

5 Robotics

Deep Reinforcement Learning Models:

- **2015: DQN (Deep Q-Network, DeepMind/Google)** - First deep RL model to achieve human-level control in Atari games.
- **2016: A3C (Asynchronous Advantage Actor-Critic, DeepMind/Google)** - A parallel asynchronous RL algorithm for efficient training.
- **2017: PPO (Proximal Policy Optimization, OpenAI)** - A stable and efficient policy gradient method widely used in practice.

- **2015: DDPG (Deep Deterministic Policy Gradient)** - An actor-critic, model-free algorithm for continuous action spaces.
- **2018: SAC (Soft Actor-Critic)** - An off-policy actor-critic algorithm aiming for maximum entropy reinforcement learning.
- **2016/2017: AlphaGo/Zero (DeepMind/Google)** - Mastered Go using deep neural networks combined with Monte Carlo Tree Search.

Sim-to-Real: Domain adaptation techniques using randomized simulations to transfer policies from simulation to physical robots.

Architectures: CNNs for vision tasks; MLPs and dedicated policy/value networks for control; common losses include Bellman error, clipped surrogate (PPO), and entropy regularization.

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